

ELITE MATHEMATICS

Pearson Edexcel International A Level

Pure 1

Adaptive Classified

Chapter 01 · Indices and Surds

With Answers

Questions with full worked solutions

- 30 pages
- Past papers up to June 2026

Indices and Surds

Your classified route

12
questions

73
marks

3
families

7
variants

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CLASSIFIED BY IDEA

Move from recognition to method, then use the working space to complete the solution.

Indices and Surds

Families and variants

F01 • Index Laws and Re-expression

V01 Algebraic Fractional and Negative Powers

V02 Re-expression Through a Named Power

V03 Single Common-base Index Equation

F02 • Power-substitution Equations

V01 Quadratic in an Exponential Power

V02 Fractional-power Equation to a Polynomial

F03 • Exact Surd Algebra

V01 Simplify, Combine and Expand Surds

V02 Rationalise a Binomial Surd Denominator

V03 Solve an Exact Linear Surd Equation

V04 Nested-surd Reciprocal Identity

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Index Laws and Re-expression

WHAT THIS FAMILY ASKS

Simplify expressions by rewriting compatible bases and applying the laws of rational indices, often to express the result in a prescribed variable or base.

VARIANT ROUTE

V01 Algebraic Fractional and Negative Powers

V02 Re-expression Through a Named Power

3 Complete Classified questions

14 total marks

2. Given that

$$a = \frac{1}{64}x^2 \qquad b = \frac{16}{\sqrt{x}}$$

express each of the following in the form kx^n where k and n are simplified constants.

(a) $a^{\frac{1}{2}}$ (1)

(b) $\frac{16}{b^3}$ (1)

(c) $\left(\frac{ab}{2}\right)^{-\frac{4}{3}}$ (2)

(Total for Question 2 is 4 marks)

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Question 1

WMA11/01 October 2023

T01 Indices and Surds

F01 Index Laws and Re-expression

V01 Algebraic Fractional and Negative Powers

1

Square root a

$$a^{1/2} = \left(\frac{1}{64}x^2\right)^{1/2} = \frac{1}{8}x.$$

2

Cube b

$$b = 16x^{-1/2}, \text{ so } b^3 = 4096x^{-3/2}.$$

3

Find 16 divided by b cubed

$$\frac{16}{b^3} = \frac{16}{4096x^{-3/2}} = \frac{1}{256}x^{3/2}.$$

4

Simplify ab over 2

$$\frac{ab}{2} = \frac{1}{2} \left(\frac{1}{64}x^2\right) (16x^{-1/2}) = \frac{1}{8}x^{3/2}.$$

5

Raise to the power -4 over 3

$$\left(\frac{1}{8}x^{3/2}\right)^{-4/3} = 8^{4/3}x^{-2} = 16x^{-2}.$$

FINAL ANSWER

$$(a) \frac{1}{8}x. \quad (b) \frac{1}{256}x^{3/2}. \quad (c) 16x^{-2}.$$

1. Given that

$$p = \frac{1}{16}x^4 \qquad q = \frac{40}{x^3}$$

express each of the following in the form kx^n where k and n are fully simplified constants.

(a) $p^{\frac{1}{2}}$ (1)

(b) $(pq)^{-1}$ (2)

(c) pq^2 (2)

(Total for Question 1 is 5 marks)

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Question 2

WMA11/01 May/June 2025

T01 Indices and Surds

F01 Index Laws and Re-expression

V01 Algebraic Fractional and Negative Powers

1

Simplify p

$$p = \frac{1}{16}x^4.$$

2

Simplify part (a)

$$p^{1/2} = \left(\frac{1}{16}x^4\right)^{1/2} = \frac{1}{4}x^2.$$

3

Simplify pq

$$pq = \frac{1}{16}x^4 \cdot \frac{40}{x^3} = \frac{5}{2}x.$$

4

Simplify part (b)

$$(pq)^{-1} = \left(\frac{5}{2}x\right)^{-1} = \frac{2}{5}x^{-1}.$$

5

Simplify part (c)

$$pq^2 = \frac{1}{16}x^4 \left(\frac{40}{x^3}\right)^2 = \frac{1600}{16}x^{-2} = 100x^{-2}.$$

FINAL ANSWER

$$(a) \frac{1}{4}x^2. \quad (b) \frac{2}{5}x^{-1}. \quad (c) 100x^{-2}.$$

2. Given $y = 3^x$, express each of the following in terms of y . Write each expression in its simplest form.

(a) 3^{3x} (1)

(b) $\frac{1}{3^{x-2}}$ (2)

(c) $\frac{81}{9^{2-3x}}$ (2)

(Total 5 marks)

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Question 3

WMA11/01 January 2020

T01 Indices and Surds

F01 Index Laws and Re-expression

V02 Re-expression Through a Named Power

1

Use $y = 3^x$ Since $y = 3^x$, powers of 3^x can be replaced by powers of y .

2

Simplify part (a)

$$3^{3x} = (3^x)^3 = y^3.$$

3

Simplify part (b)

$$\frac{1}{3^{x-2}} = 3^{2-x} = \frac{3^2}{3^x} = \frac{9}{y}.$$

4

Simplify part (c)

$$\frac{81}{9^{2-3x}} = \frac{3^4}{(3^2)^{2-3x}} = \frac{3^4}{3^{4-6x}} = 3^{6x} = (3^x)^6 = y^6.$$

FINAL ANSWER

$$(a) y^3. \quad (b) \frac{9}{y}. \quad (c) y^6.$$

Power-substitution Equations

WHAT THIS FAMILY ASKS

Replace a repeated power expression by one variable, solve the resulting polynomial and then return to the original variable with the correct domain filter.

VARIANT ROUTE

V01 Quadratic in an Exponential Power

3 Complete Classified questions

21 total marks

5.

In this question you must show all stages of your working.

Solutions relying on calculator technology are not acceptable.

(a) By substituting $p = 3^x$, show that the equation

$$3 \times 9^x + 3^{x+2} = 1 + 3^{x-1}$$

can be rewritten in the form

$$9p^2 + 26p - 3 = 0 \quad (3)$$

(b) Hence solve

$$3 \times 9^x + 3^{x+2} = 1 + 3^{x-1} \quad (3)$$

(Total for Question 5 is 6 marks)

Question 4

WMA11/01 January 2023

T01 Indices and Surds

F02 Power-substitution Equations

V01 Quadratic in an Exponential Power

1

Let $p = 3^x$ **Then $9^x = 3^{2x} = p^2$, $3^{x+2} = 9p$, and $3^{x-1} = \frac{p}{3}$.**

2

Substitute into the equation

$$3p^2 + 9p = 1 + \frac{p}{3}.$$

3

Clear the fraction**Multiplying by 3 gives $9p^2 + 27p = 3 + p$, hence $9p^2 + 26p - 3 = 0$.**

4

Solve for p

$$9p^2 + 26p - 3 = (9p - 1)(p + 3) = 0, \text{ so } p = \frac{1}{9} \text{ or } p = -3.$$

5

Reject the negative value**Since $p = 3^x > 0$, we reject $p = -3$. Therefore $3^x = \frac{1}{9} = 3^{-2}$.**

6

Find x

$$x = -2.$$

FINAL ANSWER

$$x = -2.$$

4.

In this question you must show all stages of your working.

Solutions relying on calculator technology are not acceptable.

(a) By substituting $p = 2^x$, show that the equation

$$2 \times 4^x - 2^{x+3} = 17 \times 2^{x-1} - 4$$

can be written in the form

$$4p^2 - 33p + 8 = 0 \quad (3)$$

(b) Hence solve

$$2 \times 4^x - 2^{x+3} = 17 \times 2^{x-1} - 4 \quad (3)$$

(Total for Question 4 is 6 marks)

Question 5

WMA11/01 January 2024

T01 Indices and Surds

F02 Power-substitution Equations

V01 Quadratic in an Exponential Power

1

Substitute p equals 2 to the x If $p = 2^x$, then $4^x = p^2$, $2^{x+3} = 8p$, and $2^{x-1} = \frac{p}{2}$.

2

Rewrite the equation

$$2p^2 - 8p = 17\left(\frac{p}{2}\right) - 4.$$

3

Clear the fractionMultiplying by 2 gives $4p^2 - 16p = 17p - 8$, so $4p^2 - 33p + 8 = 0$.

4

Solve the quadratic

$$4p^2 - 33p + 8 = (4p - 1)(p - 8), \text{ so } p = \frac{1}{4} \text{ or } p = 8.$$

5

Return to x

$$2^x = \frac{1}{4} = 2^{-2} \text{ gives } x = -2, \text{ and } 2^x = 8 = 2^3 \text{ gives } x = 3.$$

FINAL ANSWER

$$x = 3 \text{ or } x = -2.$$

4.

In this question you must show all stages of your working.
Solutions relying entirely on calculator technology are not acceptable.

(i) Given that

$$y = a^x \quad \text{where } a \text{ is a positive constant}$$

express, in simplest form, in terms of y and a

$$(a) \quad a^{3x+1} \quad (1)$$

$$(b) \quad \frac{5}{(3a^{1-x})^{-2}} \quad (3)$$

(ii) (a) Use the substitution $p = 9^t$ to show that the equation

$$3(3^{4t+2} + 1) = 82 \times 9^t$$

can be rewritten as

$$27p^2 - 82p + 3 = 0 \quad (2)$$

(b) Hence solve

$$3(3^{4t+2} + 1) = 82 \times 9^t \quad (3)$$

(Total for Question 4 is 9 marks)

Question 6

WMA11/01 January 2025

T01 Indices and Surds

F02 Power-substitution Equations

V01 Quadratic in an Exponential Power

1

Use y equals a to the x

$$a^{3x+1} = a(a^x)^3 = ay^3.$$

2

Rewrite the denominator

$$\frac{5}{(3a^{1-x})^{-2}} = 5(3a^{1-x})^2 = 45a^{2-2x}.$$

3

Write in terms of y

$$45a^{2-2x} = \frac{45a^2}{(a^x)^2} = \frac{45a^2}{y^2}.$$

4

Use p equals 9 to the t

$$p = 9^t = 3^{2t}, \text{ so } 3^{4t+2} = 9(3^{2t})^2 = 9p^2.$$

5

Rewrite the equation

$$3(3^{4t+2} + 1) = 82 \times 9^t \text{ becomes } 3(9p^2 + 1) = 82p.$$

6

Obtain the quadratic

$$27p^2 + 3 = 82p, \text{ so } 27p^2 - 82p + 3 = 0.$$

7

Solve for p

$$p = \frac{82 \pm 80}{54}, \text{ so } p = 3 \text{ or } p = \frac{1}{27}.$$

8

Return to t

$9^t = 3$ gives $3^{2t} = 3^1$, so $t = \frac{1}{2}$. Also $9^t = \frac{1}{27} = 3^{-3}$ gives $2t = -3$, so $t = -\frac{3}{2}$.

FINAL ANSWER

(i)(a) ay^3 . (i)(b) $\frac{45a^2}{y^2}$. (ii)(b) $t = \frac{1}{2}$ or $t = -\frac{3}{2}$.

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Exact Surd Algebra

WHAT THIS FAMILY ASKS

Simplify, rationalise and solve exact expressions involving square roots while preserving an exact requested form.

VARIANT ROUTE

V02 Rationalise a Binomial Surd Denominator

V03 Solve an Exact Linear Surd Equation

V04 Nested-surd Reciprocal Identity

6 Complete Classified questions

38 total marks

3. In this question you must show all stages of your working.

Solutions relying on calculator technology are not acceptable.

(i) Show that $\frac{\sqrt{180} - \sqrt{80}}{\sqrt{5}}$ is an integer and find its value. (2)

(ii) Simplify

$$\frac{4\sqrt{5} - 5}{7 - 3\sqrt{5}}$$

giving your answer in the form $a + b\sqrt{5}$ where a and b are rational numbers. (3)

(Total 5 marks)

Question 7

WMA11/01 May/June 2022

T01 Indices and Surds

F03 Exact Surd Algebra

V02 Rationalise a Binomial
Surd Denominator

1

Simplify the square roots

$$\sqrt{180} = 6\sqrt{5} \text{ and } \sqrt{80} = 4\sqrt{5}.$$

2

Complete part (a)

$$\frac{\sqrt{180} - \sqrt{80}}{\sqrt{5}} = \frac{6\sqrt{5} - 4\sqrt{5}}{\sqrt{5}} = 2.$$

3

Rationalise part (b)

$$\frac{4\sqrt{5} - 5}{7 - 3\sqrt{5}} \times \frac{7 + 3\sqrt{5}}{7 + 3\sqrt{5}} = \frac{(4\sqrt{5} - 5)(7 + 3\sqrt{5})}{49 - 45}.$$

4

Expand and simplify

The numerator is $28\sqrt{5} + 60 - 35 - 15\sqrt{5} = 25 + 13\sqrt{5}$, so the expression is

$$\frac{25 + 13\sqrt{5}}{4}.$$

FINAL ANSWER

$$(a) 2. \quad (b) \frac{25}{4} + \frac{13}{4}\sqrt{5}.$$

3. In this question you must show all stages of your working.

Solutions relying on calculator technology are not acceptable.

(i)

$$f(x) = (x + \sqrt{2})^2 + (3x - 5\sqrt{8})^2$$

Express $f(x)$ in the form $ax^2 + bx\sqrt{2} + c$ where a , b and c are integers to be found. (3)

(ii) Solve the equation

$$\sqrt{3}(4y - 3\sqrt{3}) = 5y + \sqrt{3}$$

giving your answer in the form $p + q\sqrt{3}$ where p and q are simplified fractions to be found. (4)

(Total 7 marks)

Question 8

WMA11/01 January 2022

T01 Indices and Surds

F03 Exact Surd Algebra

V03 Solve an Exact Linear Surd Equation

1

Expand the first square

$$(x + \sqrt{2})^2 = x^2 + 2\sqrt{2}x + 2.$$

2

Expand the second square

$$\text{Since } \sqrt{8} = 2\sqrt{2}, (3x - 5\sqrt{8})^2 = (3x - 10\sqrt{2})^2 = 9x^2 - 60\sqrt{2}x + 200.$$

3

Collect terms

$$f(x) = x^2 + 2\sqrt{2}x + 2 + 9x^2 - 60\sqrt{2}x + 200 = 10x^2 - 58\sqrt{2}x + 202.$$

4

Expand the surd equation

$$\sqrt{3}(4y - 3\sqrt{3}) = 5y + \sqrt{3} \text{ gives } 4\sqrt{3}y - 9 = 5y + \sqrt{3}.$$

5

Solve for y

$$y(4\sqrt{3} - 5) = 9 + \sqrt{3}, \text{ so } y = \frac{9 + \sqrt{3}}{4\sqrt{3} - 5}.$$

6

Rationalise the denominator

$$y = \frac{(9 + \sqrt{3})(4\sqrt{3} + 5)}{(4\sqrt{3})^2 - 5^2} = \frac{57 + 41\sqrt{3}}{23} = \frac{57}{23} + \frac{41}{23}\sqrt{3}.$$

FINAL ANSWER

$$(i) f(x) = 10x^2 - 58\sqrt{2}x + 202. \quad (ii) y = \frac{57}{23} + \frac{41}{23}\sqrt{3}.$$

3.

In this question you must show all stages of your working.

Solutions relying on calculator technology are not acceptable.

(a) Write $\frac{8 - \sqrt{15}}{2\sqrt{3} + \sqrt{5}}$ in the form $a\sqrt{3} + b\sqrt{5}$ where a and b are integers to be found. (3)

(b) Hence, or otherwise, solve

$$(x + 5\sqrt{3})\sqrt{5} = 40 - 2x\sqrt{3}$$

giving your answer in simplest form. (3)

(Total for Question 3 is 6 marks)

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Question 9

WMA11/01 October 2023

T01 Indices and Surds

F03 Exact Surd Algebra

V03 Solve an Exact Linear
Surd Equation

1

Rationalise the denominator

$$\frac{8 - \sqrt{15}}{2\sqrt{3} + \sqrt{5}} \times \frac{2\sqrt{3} - \sqrt{5}}{2\sqrt{3} - \sqrt{5}} = \frac{(8 - \sqrt{15})(2\sqrt{3} - \sqrt{5})}{12 - 5}.$$

2

Simplify the numerator

$$(8 - \sqrt{15})(2\sqrt{3} - \sqrt{5}) = 16\sqrt{3} - 8\sqrt{5} - 3\sqrt{5} + 5\sqrt{3} = 21\sqrt{3} - 14\sqrt{5}.$$

3

Divide by 7

$$\frac{21\sqrt{3} - 14\sqrt{5}}{7} = 3\sqrt{3} - 2\sqrt{5}.$$

4

Rearrange the equation

$$(x + 5\sqrt{3})\sqrt{5} = 40 - 2x\sqrt{3} \text{ gives } x(\sqrt{5} + 2\sqrt{3}) = 40 - 5\sqrt{15}.$$

5

Use the result from part (a)

$$x = \frac{40 - 5\sqrt{15}}{2\sqrt{3} + \sqrt{5}} = 5(3\sqrt{3} - 2\sqrt{5}).$$

FINAL ANSWER

$$(a) 3\sqrt{3} - 2\sqrt{5}. \quad (b) x = 15\sqrt{3} - 10\sqrt{5}.$$

2. (i) Given that $m = 2^n$, express each of the following in simplest form in terms of m .

(a) 2^{n+3} (1)

(b) 16^{3n} (2)

(ii) **In this question you must show all stages of your working.**

Solutions relying on calculator technology are not acceptable.

Solve the equation

$$x\sqrt{3} - 3 = x + \sqrt{3}$$

giving your answer in the form $p + q\sqrt{3}$ where p and q are integers. (3)

(Total for Question 2 is 6 marks)

Question 10

WMA11/01 May/June 2024

T01 Indices and Surds

F03 Exact Surd Algebra

V03 Solve an Exact Linear
Surd Equation

1

Use m equals 2 to the n

$$2^{n+3} = 2^n \times 2^3 = 8m.$$

2

Rewrite 16 as a power of 2

$$16^{3n} = (2^4)^{3n} = 2^{12n} = (2^n)^{12} = m^{12}.$$

3

Collect the x terms

$$x\sqrt{3} - 3 = x + \sqrt{3} \text{ gives } x(\sqrt{3} - 1) = 3 + \sqrt{3}.$$

4

Rationalise

$$x = \frac{3 + \sqrt{3}}{\sqrt{3} - 1} \times \frac{\sqrt{3} + 1}{\sqrt{3} + 1} = \frac{(3 + \sqrt{3})(\sqrt{3} + 1)}{2}.$$

5

Simplify

$$x = \frac{4\sqrt{3} + 6}{2} = 3 + 2\sqrt{3}.$$

FINAL ANSWER

$$(i)(a) 8m. \quad (i)(b) m^{12}. \quad (ii) x = 3 + 2\sqrt{3}.$$

2.

In this question you must show all stages of your working.

Solutions relying on calculator technology are not acceptable.

(i) Simplify fully

$$\frac{3y^3(2x^4)^3}{4x^2y^4} \quad (3)$$

(ii) Find the exact value of a such that

$$\frac{16}{\sqrt{3} + 1} = a\sqrt{27} + 4$$

Write your answer in the form $p\sqrt{3} + q$ where p and q are fully simplified rational constants.

(4)

(Total for Question 2 is 7 marks)

Question 11

WMA11/01 October 2024

T01 Indices and Surds

F03 Exact Surd Algebra

V03 Solve an Exact Linear
Surd Equation

1

Simplify the power bracket

$$(2x^4)^3 = 8x^{12}.$$

2

Simplify the algebraic fraction

$$\frac{3y^3(2x^4)^3}{4x^2y^4} = \frac{3y^3(8x^{12})}{4x^2y^4} = \frac{6x^{10}}{y}.$$

3

Rationalise the left side

$$\frac{16}{\sqrt{3}+1} \times \frac{\sqrt{3}-1}{\sqrt{3}-1} = 8\sqrt{3}-8.$$

4

Equate the surd parts

$$8\sqrt{3}-8 = a\sqrt{27}+4 = 3a\sqrt{3}+4.$$

5

Solve for a

$$3a\sqrt{3} = 8\sqrt{3}-12, \text{ so } a = \frac{8\sqrt{3}-12}{3\sqrt{3}} = \frac{8}{3} - \frac{4}{3}\sqrt{3}.$$

FINAL ANSWER

$$(i) \frac{6x^{10}}{y}. \quad (ii) a = -\frac{4}{3}\sqrt{3} + \frac{8}{3}.$$

6.

In this question you must show all stages of your working.

Solutions relying on calculator technology are not acceptable.

(a) Expand and simplify

$$\left(r - \frac{1}{r}\right)^2$$

(2)

(b) Express $\frac{1}{3 + 2\sqrt{2}}$ in the form $p + q\sqrt{2}$ where p and q are integers.

(2)

(c) Use the results of parts (a) and (b), or otherwise, to show that

$$\sqrt{3 + 2\sqrt{2}} - \frac{1}{\sqrt{3 + 2\sqrt{2}}} = 2$$

(3)

(Total for Question 6 is 7 marks)

Question 12

WMA11/01 May/June 2023

T01 Indices and Surds

F03 Exact Surd Algebra

V04 Nested-surd Reciprocal Identity

1

Expand the square

$$\left(r - \frac{1}{r}\right)^2 = r^2 - 2 + \frac{1}{r^2}.$$

2

Rationalise the denominator

$$\frac{1}{3 + 2\sqrt{2}} \times \frac{3 - 2\sqrt{2}}{3 - 2\sqrt{2}} = \frac{3 - 2\sqrt{2}}{9 - 8} = 3 - 2\sqrt{2}.$$

3

Let r be the square root

Let $r = \sqrt{3 + 2\sqrt{2}}$. Then $r^2 = 3 + 2\sqrt{2}$ and $\frac{1}{r^2} = 3 - 2\sqrt{2}$.

4

Use part (a)

$$\left(r - \frac{1}{r}\right)^2 = r^2 - 2 + \frac{1}{r^2} = (3 + 2\sqrt{2}) - 2 + (3 - 2\sqrt{2}) = 4.$$

5

Take the positive square root

Since $r > 1$, $r - \frac{1}{r} > 0$, so $r - \frac{1}{r} = 2$.

FINAL ANSWER

(a) $r^2 - 2 + \frac{1}{r^2}$. (b) $3 - 2\sqrt{2}$. (c) The expression equals 2.